JK TYRE & INDUSTRIES LTD.

4RC Production Line Changeover Analytics

Technical Documentation System Features & Working Logic

---

1. System Overview

The 4RC Production Line Changeover Analytics Dashboard is a web-based industrial monitoring system developed for JK Tyre & Industries Ltd. It is designed to track, analyse, and report changeover events on the 4RC tire production line, enabling plant managers and floor workers to monitor setup efficiency, identify time overruns, and take corrective actions in a structured workflow.

Attribute	Details
Project Name	4RC Setup Analytics Dashboard
Client	JK Tyre & Industries Ltd.
Backend Framework	Django (Python) + Django REST Framework
Frontend Framework	React 18 + Vite
Database	MySQL
Task Queue	Celery + Redis
Report Generation	PDF (server-side) & Excel
Authentication	Token-based (email/password)

1.1 Prerequisites

- Python 3.9+
- Node.js 18+
- MySQL 8.0+
- Redis (for Celery broker)

---

2. System Features

The system is organized around core features, each serving a distinct operational purpose for plant personnel.

2.1 4RC Setup Summary Dashboard

The central interface displays all changeover events within a user-specified date range and shift. The dashboard supports three viewing modes: Setup Details (tabular), Charts (graphical), and Setup Summary (aggregated).

Filter Controls :

- **Shift selector:** All Shifts, Shift A (07:00–15:00), Shift B (15:00–23:00), Shift C (23:00–07:00)
- **Recipe/machine selector:** Filter by specific machine code (e.g., CP 034 HI) or view all
- **Date range picker:** Custom start–end date selection - GET button: Fetches data for the selected filter combination

Setup Summary Table Columns

Column	Description
Type of Style	Changeover category (e.g., Fabric to Fabric, Steel to Steel)
STD Time (min)	Standard expected time for this changeover type from Standard Time Master
ACT Time (min)	Actual total setup time recorded (Static S/U + Ramp Up)
Static S/U (min)	Time the line was completely stopped (speed = 0)
Ramp Up (min)	Time taken for the line to accelerate from 0 to near-target speed
Setup Count	Number of changeover events of this type in the selected period
Over Shoot (min)	ACT Time minus STD Time — highlighted red if positive (exceeded standard)
Action	Show Details button to drill into individual changeover records

2.2 CHARTS AND GRAPHS

2.2.1 CHART VISUALIZATION The system includes multiple charts to visualize production setup performance. These charts help users analyze trends, compare actual and standard times, and identify causes of delays effectively.

2.2.2 LINE CHART – ACTUAL vs STANDARD TIME - Line chart shows comparison between actual time and standard time for each setup - X-axis represents setup sequence and Y-axis represents time in minutes - Two lines are plotted: one for actual time and one for standard time - A highlighted area appears when actual time is greater than standard time, indicating delay - Helps in identifying performance gaps and trends over time

2.2.3 BAR CHART – OVERSHOOT CATEGORIES - Bar chart displays overshoot count based on categories like mechanical, electrical, operation, and others - X-axis shows categories and Y-axis shows number of occurrences - Each bar represents how many times delay occurred in that category - Helps in identifying which category contributes most to delays

2.2.4 BAR CHART – OVERSHOOT REASONS - Another bar chart shows detailed overshoot reasons within each category - X-axis represents specific reasons and Y-axis shows frequency - Bars are color-coded based on category - Helps in identifying exact causes of delays and recurring issues

2.3 Changeover Detail Drill-Down

Clicking Show Details on any row opens a modal showing every individual changeover event under that type. Each row displays: - Material (recipe code used for the new run) - STD Time, ACT Time, Static S/U, Ramp Up, Over Shoot for the specific event - Start Time: the timestamp when setup commenced (Point B) - Category dropdown: Mechanical, Electrical, Operational, Material, Other, or None (Admin adds manually using SQL) - Reason text field: free-text explanation entered by the worker - Update / Save button to persist the reason

Overshoot values are shown in red when positive (actual exceeded standard) and in green/blue when negative (setup completed faster than standard).

2.4 Recipe Master

The Recipe Master defines the mapping between recipe codes, SAP codes, recipe types, and target machine speeds. This data is foundational — without a valid recipe entry, changeover events cannot be processed correctly.

Automated BOM Sync To ensure strict data integrity, the system features an automated integration with the external ERP/BOM database (the bom table, represented by the MaterialMaster model). A background task (sync_recipe_master_from_bom) automatically pulls and syncs new recipes designated for the 4RC equipment, populating the recipe_code and sap_code directly from the source.

Managing Recipes Because the creation of new recipes is completely automated via the BOM sync, the ability to manually add new rows or bulk-upload recipes has been disabled. Through the dashboard, managers are only permitted to update two specific fields on existing, synced recipes:

- Target Speed: (m/min)
- Recipe Type: (e.g., Fabric or Steel)

Export The *Download Currently Active Recipes* button exports the full current recipe list to a spreadsheet for offline reference or audit purposes.

2.5 Standard Time Master

The Standard Time Master stores the expected (benchmark) setup durations for each changeover type. These values are used to compute the Overshoot figure shown throughout the dashboard.

Changeover Type (Key)	Example Standard Time
Steel to Steel	67 minutes
Steel to Fabric	30 minutes
Fabric to Steel	61 minutes

Changeover Type (Key)	Example Standard Time
Fabric to Fabric	10 minutes
STEEL-STEEL ONLY COMPOUND CHANGE	25 minutes

2.6 Overshoot Tracking & Alerting

Every processed changeover event is automatically compared against its Standard Time. The resulting Overshoot value is displayed both in the summary table and in detail drill-downs.

Alert Thresholds - Any positive overshoot is highlighted in red in the UI. - When actual time exceeds 2× the standard time, a critical alert signal is triggered server-side (Django signal handler), which can be connected to notification mechanisms. - The Header component polls the backend every 2 minutes to surface any new pending correction requests or critical alerts to the manager in real time.

Missing Recipe Warnings If the automated processor encounters a recipe code in the raw sensor data that has no entry in the Recipe Master, a warning is raised. Managers can view all unresolved missing-recipe warnings via the dedicated warnings panel in the header navigation.

2.7 5-Hour Correction Window & Approval Workflow

To ensure data integrity while allowing legitimate corrections, the system enforces a time-based editing policy.

Correction Policy - **Within 5 hours of a changeover event:** Worker can freely update the overshoot category and reason. Changes are saved immediately. - **After 5 hours:** The record is locked. The worker can still submit a correction, but it creates a Pending Correction Request rather than updating the record directly. - **Manager review:** The manager sees all pending requests, reviews the proposed change, and either Approves (data is updated) or Rejects (data remains unchanged). - **Threshold constant:** 5 hours is a configurable constant in the backend settings.

2.8 User Authentication & Role Management

The system enforces role-based access control. Three user roles are defined, each with a distinct permission set.

Role	Capabilities	Restrictions
Admin	Full system access — manage users, all data, all settings	None
Manager	Upload recipes, manage standard times, approve/reject correction requests, view all warnings and reports	Cannot perform admin-level user management
Worker	View dashboard, update overshoot reason/category	Cannot upload recipes

Role	Capabilities	Restrictions
	within 5-hour window, submit correction requests after the window	

Authentication is token-based. Upon successful login via email and password, the server returns a session token that is stored in the browser's localStorage and attached to all subsequent API requests.

2.9 PDF & Excel Report Generation

2.9.1 Pdf Download Feature The PDF download feature allows users to export production reports in a structured and printable format. When the user clicks the download option, a request is sent to the backend, where the required data is retrieved and converted into a PDF file. The generated file is then returned to the frontend and downloaded automatically.

2.9.2 Excel Download Feature The Excel download feature enables users to export production data in spreadsheet format for analysis. The frontend sends a request to the backend, which fetches the data and converts it into an Excel (.xlsx) file. The file is then sent back and downloaded by the user.

2.10 Date & Shift Filtering

All data views respect the active date range and shift filter. Shift boundaries are defined as: - Shift A: 07:00 – 15:00 - Shift B: 15:00 – 23:00 - Shift C: 23:00 – 07:00 (crosses midnight) Selecting All Shifts returns data across all three shifts for the chosen date range.

---

3. Working Logic

This section describes the complete data pipeline — from raw sensor data collected on the production line to the final analytics displayed on the dashboard. This is the core intelligence of the system.

3.1 Raw Data Collection

The 4RC production line is equipped with sensors that record machine speed (in m/min) and the active recipe code at every second. This data is stored directly into the FRC table (also referred to as RawLineData) in the MySQL database. Each row contains at minimum: a timestamp, the current line speed, and the active recipe code.

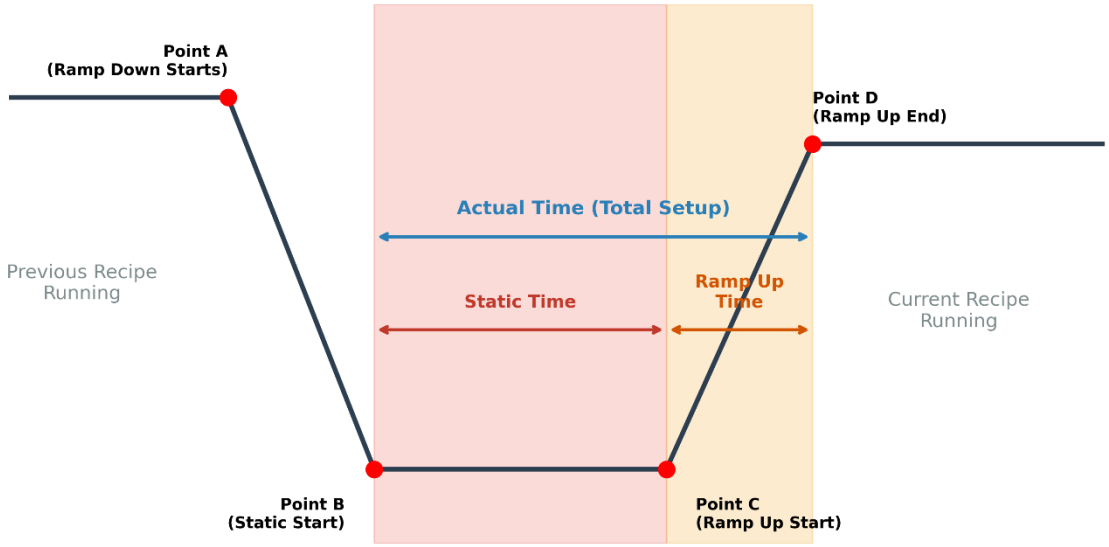
3.2 Automated Changeover Processing (Celery Task)

A Celery background worker runs automatically every 5 minutes. It scans the FRC table for unprocessed data and applies the changeover detection algorithm described below. Redis is used as the message broker for Celery.

The algorithm identifies when a changeover has occurred by detecting the four key points — A, B, C, and D — on the line speed signal. These points mark the phases of a changeover event.

3.3 Line Speed Graph — The Four Key Points

The line speed graph behaves in distinct phases during a changeover. The four key points are calculated using precise logical thresholds:



Actual Line Speed Graph

Point A — Ramp Down Starts Point A is the moment the outgoing recipe begins decelerating. It is identified when the machine drops out of its operational speed zone. Specifically, the system marks the last reading where the machine speed was equal to or greater than **90% of the outgoing recipe's target speed** before falling below that threshold.

Point B — Machine Stops (Setup Begins) Point B marks the exact moment the machine comes to a complete standstill (LS DROP) and active setup work begins. To avoid false positives from transient sensor noise in the previous recipe, it is detected algorithmically by looking for at least two consecutive sustained speed readings (LS >= 10 m/min) that are definitively followed by a reading of 0 m/min. (As a secondary fallback, the system may also use a 15-sample rolling window of LS >= 4 leading to a zero).

Point C — Ramp Up Begins Point C is identified when the line speed begins to increase again from zero. The system scans the zero-speed period and identifies Point C as the first instance of a sustained machine movement. To filter out transient noise, the algorithm requires a rolling window of 15 consecutive readings where the speed is consistently >= 4 m/min, marking the start of that window as Point C to indicate the core setup work is over and true acceleration has begun.

Point D — Setup Complete (Target Speed Reached) Point D marks the end of the changeover. The algorithm uses a sliding window of 10 consecutive speed samples. When **7 out of 10 values** within this window consistently reach **90% or more of the new recipe's target speed**, Point D is confirmed at the first valid point in that window.

3.4 Time Calculations

Once the four key points are identified, three critical time metrics are derived using absolute time and Area Under Curve (AUC) methods:

Metric	Formula / Logic	Description
Static Setup Time	Time(C) − Time(B)	Duration the line was fully stopped (0 m/min). Represents the core hands-on setup work.
Ramp Up Time	Time(D) − Time(C)	Total elapsed time taken for the line to accelerate from rest to production speed.
Actual Total Setup Time	Static S/U + Ramp Up Time	Complete elapsed time from machine stop to production speed restored.

*Note: If the machine never drops to 0 speed, or never reaches the 90% target threshold, the system flags the batch with specific **Remarks** explaining why the calculations were aborted.*

3.5 Overshoot Calculation

Once the Actual Total Setup Time is known, it is compared against the Standard Time for that changeover type retrieved from the Standard Time Master:

Overshoot (min) = Actual Total Setup Time − Standard Time

- Positive overshoot → Setup exceeded the standard → Highlighted RED
- Negative overshoot → Setup faster than standard → Shown in green/neutral
- Critical Alert: If Actual Time > 2 × Standard Time → Signal fired server-side

3.6 Data Storage — ChangeoverSummary

After calculations, the system writes a new record to the ChangeoverSummary table. This record stores all computed values and serves as the primary data source for the dashboard. Key fields include: -

- Material (incoming recipe code)
- Changeover type string (e.g., ‘Fabric to Fabric’)
- Static setup time, ramp up time, actual total time
- Ramp up time loss, Ramp down time loss ,Standard time (snapshot at time of processing)
- overshoot value
- Start timestamp (Point B)
- Shift (A, B, or C) derived from the start timestamp

- Overshoot category and reason (initially blank, filled in by worker)
- Remarks (auto-generated explanations if calculations fail) - Locked status and correction request linkage

3.7 System Status Tags & Flags (Newly Added)

To ensure reliable processing and auditing, the system applies robust status tags at various stages:

Raw Data Processing Tags (RawLineData.status): -

- **SUCCESS:** Complete valid changeover perfectly succeeded (both incoming and outgoing transitions verified).
- **IGNORED_LINE_PASS:** Records associated with a LINE PASS recipe are safely ignored.
- **SKIPPED_NO_TARGET_SPEED:** Skipped because the target speed is missing in the Recipe Master.
- **FAILED_OPERATIONAL_SPEED:** Failed because physical requirements were not met (e.g., the machine never reached the 90% operational threshold speed).

Correction Request Tags (ChangeoverCorrectionRequest.status):

- PENDING: Correction requested by worker (after 5h), awaiting manager review.
- APPROVED: Manager approved request, updating the main record.
- REJECTED: Manager rejected request, leaving original data intact.

3.8 Worker Update & Correction Request Workflow

After a changeover record is created, workers review it and assign the root cause of any overshoot.

Correction Workflow Timeline

Step 1 — Setup event processed → ChangeoverSummary record created (category and reason are blank)

Step 2 — Worker opens the detail drill-down within 5 hours → Selects category (e.g., Electrical), enters reason (e.g., ‘power fluctuation’) → Saves directly

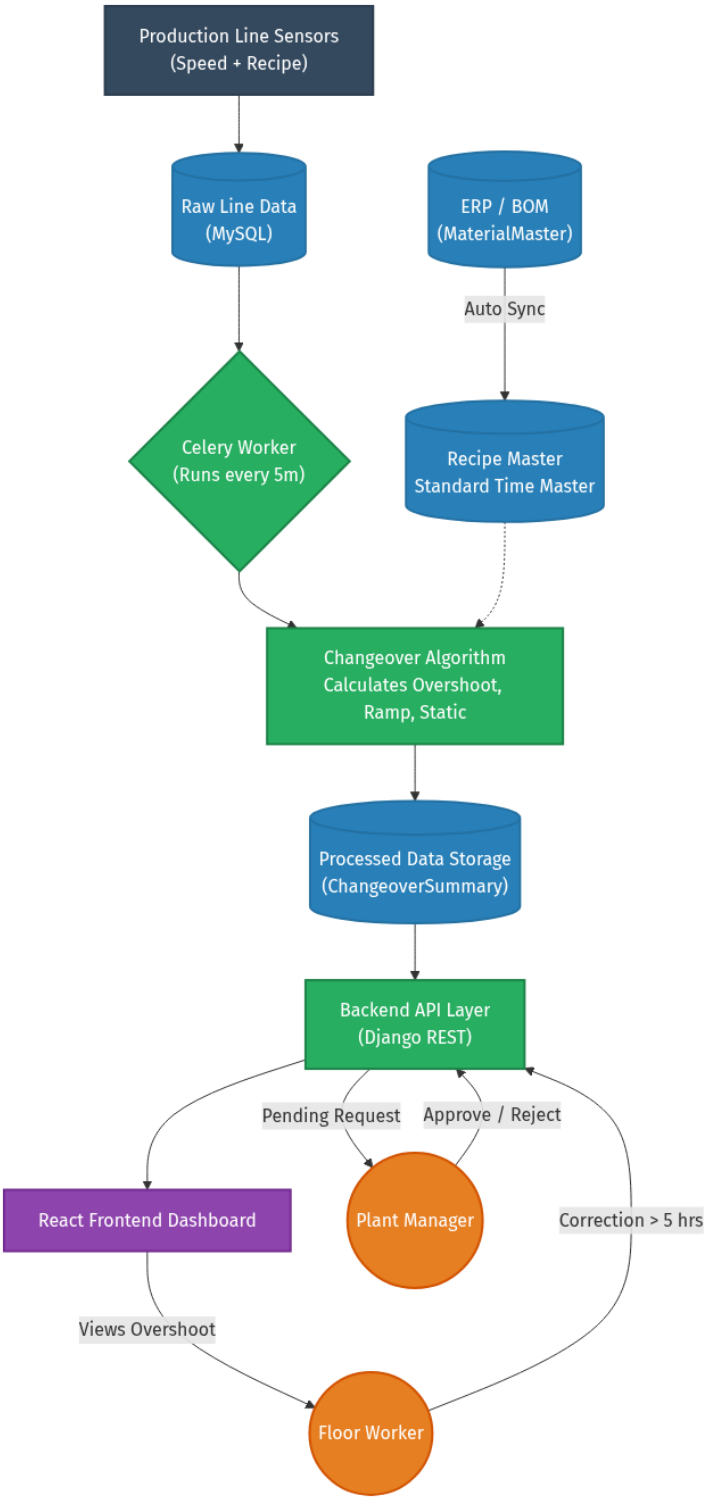
Step 3 — After 5 hours → Worker can still submit an update, but it creates a pending Correction Request instead of saving directly

Step 4 — Manager sees the pending request in the corrections panel (polled every 2 minutes by the header)

Step 5 — Manager approves → Record is updated with the new category/reason

Step 6 — Manager rejects → Record remains unchanged, worker is notified

3.9 Complete End-to-End Data Flow



4. API Reference

4.1 Authentication

Method	Endpoint	Description	Access
POST	/api/auth/signup/	Register a new user with role assignment	Public
POST	/api/auth/login/	Login — returns authentication token	Public

4.2 Recipe Management

Method	Endpoint	Description	Access
GET	/api/recipe-master/	Lists all recipes	Any Authenticated user
PATCH	/api/recipe-master/<str:pk>/	To update target speed or recipe type	Any Authenticated user
GET	/api/recipe-export-master/	To download recipes in various formats such as json , xls , csv using ?format = excel	Any Authenticated user
GET	/api/missing-recipes-warning/	Get list of unresolved missing recipe warnings	Admin

4.3 Changeover Data

Method	Endpoint	Description	Access
GET	/api/changeover-stats/	Fetch dashboard data. Supports query params: date_from, date_to, shift, recipe	Any authenticated user
PATCH	/api/changeover/update/<id>/	Update overshoot category and reason for a specific event	Any authenticated user
GET	/api/overshoot-options/	Retrieves all available overshoot categories	Any authenticated user

4.4 Correction Requests

Method	Endpoint	Description	Access
GET	/api/correction-	List all pending	Manager

Method	Endpoint	Description	Access
	requests/	correction requests	
POST	/api/correction-requests/<id>/action/	Approve or reject a correction request	Manager

4.5 Standard Time & Reports

Method	Endpoint	Description	Access
GET	/api/standard-time/	List all standard time entries	Any Authenticated user
GET	/api/reports/summary/	Generate and download PDF/Excel report for current filters	Any Authenticated user
GET	/api/changeover-export/	Retrieve the changeover summary table data as excel	Any Authenticated user